

Spectral degeneration can identify a direction, High-frequency-blind finite scalar filtering cannot replace all-frequency coverage

I divide the rank-deficient regime left by R0.70N into coercive, near-plane, and near-line branches. Only the near-line branch has the correct algebraic form of two transverse vorticity components; but an exact periodic shear shows that a finite scalar filter family whose high-frequency response tends to zero cannot uniformly reconstruct the unfiltered critical $L_t^4 L_x^2$ norm from this observed residual.

Status · R0.700 exact certificate, independent internal review PASS, and formal figure complete

Direct reconstruction from high-frequency-blind finite filters closed

Version v0.700 · 2026-08-25
Exact producer: all 5 groups passed
Focused tests: 9 / 9
Full regression: 609 / 609
Next gate: all-frequency lower frame and variable-direction commutator

00 / EXACT CONCLUSION

Two spectral residuals characterize the best plane and best line exactly

Let the eigenvalues of $Q = \int V \otimes V d\mu$ satisfy $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 0$, and write $E = \text{tr } Q$. Rayleigh–Ritz gives the following unnormalized identities; the normalized trichotomy below is used only where $E > 0$.

$$\lambda_3 = \min_{|n|=1} \int |n \cdot V|^2 d\mu, \quad \lambda_2 + \lambda_3 = \min_{|\ell|=1} \int |P_{\ell^\perp} V|^2 d\mu.$$

The first formula measures the absolute energy by which the data depart from the best plane; the second measures the absolute energy by which they depart from the best line. A normalized ratio describes only geometric shape and cannot replace the absolute residual.

Core conclusion of this stage. The spectral stratification itself is valid, but a finite scalar-filter residual whose high-frequency response tends to zero cannot by itself uniformly control the critical norm of the unfiltered transverse vorticity. This conclusion rules out one direct bridge; it is not a Navier–Stokes regularity theorem.

01 / SPECTRAL TRICHOTOMY

The three regions can be made disjoint and exhaustive, each with the correct spectral gap

Fix $0 < 2\delta < \eta < 1/2$. I use the following precedence-based definitions:

$$\begin{aligned} \mathbf{C}_\delta &: \lambda_3/E \geq \delta, \\ \mathbf{L}_{\delta,\eta} &: \lambda_3/E < \delta, \quad (\lambda_2 + \lambda_3)/E \leq \eta, \\ \mathbf{P}_{\delta,\eta} &: \lambda_3/E < \delta, \quad (\lambda_2 + \lambda_3)/E > \eta. \end{aligned}$$

They represent quantitative coercivity, near-line behavior, and near-plane behavior, respectively, and they do not overlap. The spectral ordering immediately gives

$$\mathbf{L}_{\delta,\eta}: \quad \lambda_1 - \lambda_2 \geq (1 - 2\eta)E, \quad \mathbf{P}_{\delta,\eta}: \quad \lambda_2 - \lambda_3 > (\eta - 2\delta)E.$$

These lower bounds make the corresponding principal-line or normal projector algebraically stable, but they do not automatically provide regularity of its spatial variation.

02 / SPECTRAL EVOLUTION

The spectral gap controls projector derivatives, but the source terms still require term-by-term estimates

For the symmetric-matrix ledger

$$\dot{Q} = \Sigma Q + Q \Sigma + F, \quad \Sigma = \Sigma^\top, \quad F = F^\top,$$

In the simple-spectrum region, the exact formulas in the eigenbasis are

$$\dot{\lambda}_a = 2\lambda_a \sigma_{aa} + f_{aa}, \quad e_b^\top \dot{e}_a = \frac{(\lambda_a + \lambda_b) \sigma_{ba} + f_{ba}}{\lambda_a - \lambda_b}.$$

The same perturbation formula applies to spatial derivatives. The near-line principal projector P_1 satisfies

$$\|\partial_i P_1\|_F \leq \frac{\|\partial_i Q\|_F}{\lambda_1 - \lambda_2}.$$

Thus the spectral gap supplies only the denominator. A Miller-type direction condition still requires control of $\nabla Q/E$, consistent orientation or a direct projector formulation, and closure of the Navier–Stokes source terms in F .

03 / SMOOTH QUANTITATIVE OBSTRUCTION

The observed residual tends to zero while the unfiltered critical norm remains unchanged

Let J be a finite index set, and let T_j be real-preserving scalar Fourier filters acting componentwise. Fix nonnegative weights $w_j \geq 0$, and denote the total response by

$$A(k) = \sum_j w_j |m_j(k)|^2.$$

Assume $A(e_2) > 0$, and suppose there are integers $N_q \rightarrow \infty$, $N_q \geq 2$, such that $A(N_q e_2) \rightarrow 0$. On the normalized three-dimensional torus, take $\kappa > 1$, set $a_N^2 A(e_2) = 1 + \kappa N A(N e_2)$, and define

$$u_N = a_N e^{-\nu t} \sin y e_1 + N^{-1/2} e^{-\nu N^2 t} \sin(Ny) e_3.$$

This is an exact smooth, unforced, global shear solution: its divergence is zero, $(u_N \cdot \nabla) u_N = 0$, and it satisfies the heat equation. The principal line of the observed covariance is always e_3 . If $r_N = \lambda_2 + \lambda_3$, then

$$\|r_N\|_{L_t^2} = \frac{A(N e_2)}{4\sqrt{\nu}} \rightarrow 0, \quad \|P_{e_3^\perp} \omega_N\|_{L_t^4 L_x^2} = \frac{1}{2\nu^{1/4}}.$$

Thus no uniform estimate depending only on this residual exists, nor does any uniform reconstruction modulus continuous at zero and satisfying $\Phi(0) = 0$. The hypothesis $A(N_q e_2) \rightarrow 0$ must be retained here: being a finite filter family is not enough, and an identity observation is outside this no-go theorem.

04 / TIME AND ENDPOINT BOUNDARY

A finite time interval gives a sharp ratio; compact-band qualitative failure occurs at the rough initial endpoint

For any fixed $T > 0$, let $\theta_{N,T} = 1 - e^{-4\nu N^2 T}$. When $A(N e_2) > 0$, the smooth family above satisfies

$$\frac{\|P_{e_3^\perp}\omega_N\|_{L^4(0,T;L^2)}^2}{\|r_N\|_{L^2(0,T)}} = \frac{1}{A(Ne_2)}.$$

The instability is concentrated in an initial layer of width about N^{-2} . It therefore does not exclude an estimate on a fixed positively delayed interval $[\tau, \infty)$ whose constant is allowed to depend on τ .

For compact-band filters, infinitely many hidden modes with $n_k = 2^k$ can also be superposed to obtain a Leray–Hopf shear solution: the observed residual is identically zero, but

$$\frac{1}{8\nu} \sum_{k,l} \frac{n_k n_l}{n_k^2 + n_l^2} \geq \frac{1}{16\nu} \sum_k 1 = \infty.$$

This initial datum belongs to $L^2 \setminus H^1$, and the solution is smooth for every $t > 0$. It gives qualitative reconstruction failure at the initial endpoint and does not conflict with directional criteria that assume H^1 initial data.

05 / EXACT POSITIVE REPLACEMENT

The correct condition for a fixed projection is an all-frequency lower frame

For a fixed orthogonal projection P , define

$$R_P(f) = \sum_j w_j \|T_j P f\|_2^2 = \sum_k A(k) |P \widehat{f}(k)|^2.$$

Fix a frequency set Λ , and let f have Fourier support in Λ . The inequality $a_0 \|P f\|_2^2 \leq R_P(f)$ holds for every such f if and only if the following bound holds whenever $\text{Ran } P$ contains an admissible Fourier coefficient at $k \in \Lambda$

$$A(k) \geq a_0 > 0.$$

For mean-zero f , the corresponding $\dot{H}^{-1/2}$ reconstruction holds if and only if $A(k) \geq a_0 |k|^{-1}$ at every nonzero admissible frequency. In particular, if the L^2 lower-frame condition covers every nonzero admissible vorticity frequency and the absolute residual for fixed P satisfies $R_P \in L_t^2$, then

$$\|P\omega\|_{L^4(0,T;L^2)}^4 \leq a_0^{-2} \|R_P\|_{L^2(0,T)}^2.$$

This is a genuine but conditional linear bridge. A finite set of adjacent scales usually does not satisfy the all-frequency lower bound; only a complete Littlewood–Paley square function can potentially satisfy the corresponding coverage condition.

Once the principal direction varies in space and time, the commutator becomes the new core term

If $P = P_{v^\perp}(x, t)$, scalar filtering no longer commutes with multiplication by the projection:

$$T_j(P\omega) = PT_j\omega + [T_j, P]\omega.$$

A viable route must at least provide all-frequency coverage, L_t^2 control of the absolute residual, a uniform principal spectral gap, control of $\nabla Q/E$, an oriented direction or a projector formulation, and a commutator estimate summable across scales. Without every one of these ingredients, the filtered covariance cannot be inserted directly into an existing continuation criterion.

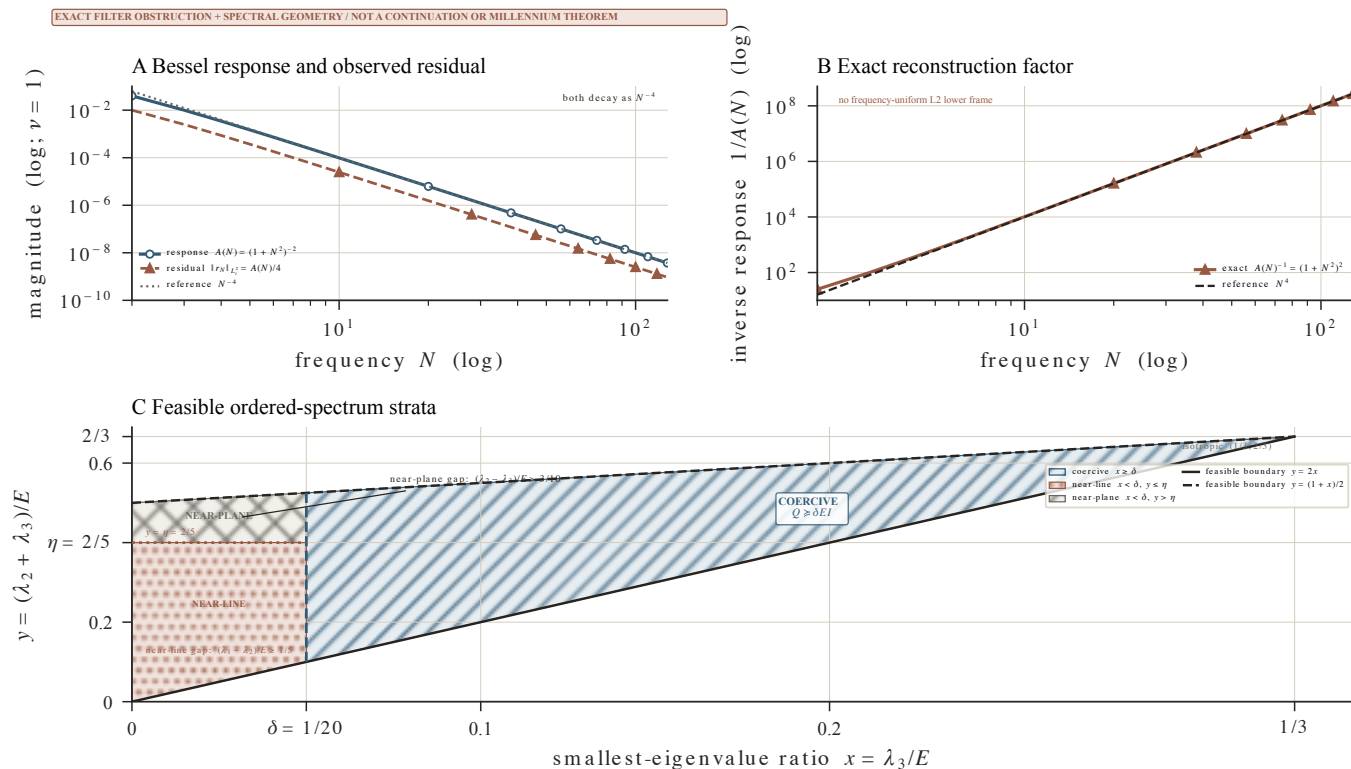
Chae–Choe uses two unfiltered vorticity components relative to one fixed global direction; Miller uses a spatially Lipschitz variable unit direction and the critical $L_t^4 L_x^2$ condition; Constantin–Fefferman uses pointwise directional coherence on the high-vorticity region. An averaged covariance cannot replace these hypotheses without a new bridge.

The exact counterexample here lies on $\mathbb{T}^3$, whereas Miller's original theorem is stated on $\mathbb{R}^3$. To use it as a downstream theorem, a periodic version must first be cited or proved, or the entire bridge must be rebuilt on the whole space.

High-frequency response, reconstruction instability, and spectral regions

Filtered rank diagnostics: reconstruction loss and feasible spectral strata

exact Bessel response · frequency-uniform lower-frame loss · ordered covariance geometry



filtered near-rank evidence reaches an unfiltered continuation criterion only after an all-frequency lower frame and a controlled direction ledger

Figure Ro.70O. A: for the Bessel filter, both the response $A(N) = (1 + N^2)^{-2}$ and the observed residual $A(N)/4$ decay like N^{-4} . B: the sharp reconstruction ratio $1/A(N) = (1 + N^2)^2$ diverges. C: in the feasible region with $x = \lambda_3/E$ and $y = (\lambda_2 + \lambda_3)/E$, $\delta = 1/20$ and $\eta = 2/5$ give disjoint coercive, near-line, and near-plane regions. Every curve comes from a closed formula, not DNS.

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This step separates geometric proximity from analytic reconstructibility

The near-line spectrum does extract a geometric quantity analogous to two transverse vorticity components; this is the valid part retained from the preceding multiscale-frame route. But finite smoothing observations lose high frequencies, so even a very small spectral residual does not automatically control the unfiltered critical norm.

The main value of this result is route selection. I can now stop looking for a uniform inequality that directly derives a critical directional condition from a finite-filter residual, and turn to two auditable gaps: an all-frequency lower frame and the variable-

direction commutator. The result also gives a positive theorem that states exactly what information must be restored.

This is not a substantive answer to the Millennium problem. It neither narrows the set of all possible singularities nor rules out singularity; it closes only one natural but incorrect inference in the current covariance-observation scheme.

09 / NEXT ROUTE

In Ro.70P, I will first build the pure harmonic-analysis bridge, then test whether NSE can propagate its hypotheses

In the next stage, I will first prove a conditional theorem: an all-frequency scalar frame, a uniform principal spectral gap, L_t^2 control of the absolute near-line residual, and a scale-summable $[T_j, P_{v^\perp}]\omega$ estimate imply that the unfiltered transverse vorticity belongs to $L_t^4 L_x^2$.

I will then test term by term whether these hypotheses can be propagated from the local covariance PDE ledger, instead of merely assuming the target critical norm in different notation. If the commutator estimate inevitably uses the target norm circularly, this route should stop. There is no need to start large-scale DNS before this algebraic and harmonic-analysis gate is complete.

10 / CLAIM BOUNDARY

What is closed is the uniform direct bridge for high-frequency-blind observations, not the direction-regularity problem

PROVED Spectral trichotomy and projector gaps The three regions are disjoint and exhaustive, and the best-plane and best-line residuals are exact.	PROVED Smooth quantitative reconstruction obstruction The observed residual tends to zero while the unfiltered critical transverse norm remains constant.
PROVED Fixed-projection lower-frame theorem An all-frequency response lower bound gives the correct conditional $L_t^4 L_x^2$ bridge.	OPEN Variable direction and NSE propagation The commutator, spatial regularity, orientation, and source-term closure remain unresolved.

The smooth sequence rules out only uniform quantitative reconstruction; the compact-band example gives qualitative failure only at a rough initial endpoint; and the present calculation does not exclude an estimate after a fixed positive delay.

This release does not obtain a new continuation criterion, construct a singularity, or prove global smoothness, and it does not solve the Millennium problem.

The exact producer, primary-literature audit, independent review, and formal figure package are all archived

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The producer certifies five groups of finite symbolic, rational, Fourier, and integral identities. Rayleigh–Ritz for arbitrary measures, the infinite dyadic series, and necessity and sufficiency of the lower frame for arbitrary filter symbols are established analytically. The independent review reran the producer and obtained the same SHA-256 as the archived file.